St. Philomena's College (Autonomous), Mysore
PG Department of Chemistry
Question Bank (Revised Curriculum 2018 onwards)
Second Year – Third Semester (2019 -21 Batch)
Course Title (Paper Title): Biomolecules. QP Code: 55206

Unit	SI. No	Questions	Marks
1	1	How is CNBr useful in the structural elucidation of peptides.	02
1	2	Write the structure of Val.Cys.H.Gly.	02
1	3	What is renaturing of proteins? Give an example.	02
1	4	What is Merrifield resin?	02
1	5	What is denaturation of proteins?	02
1	6	Give the name and structures of any two protecting groups for amino group protection.	02
1	7	Expain briefly the use of HOBt in suppression of racemization during peptide synthesis.	02
1	8	Write the name and structures of bases present in RNA.	02
1	9	What is isoelectric point? Give an examples.	02
1	10	Explain briefly protection and deprotection of amino groups in peptide synthesis.	02
1	11	Write the biological importance of insulin.	02
1	12	What is renature and denature of proteins with examples.	02
1	13	Write the biological importance of vasopressin	02
1	14	What are essential amino acids? Write any two essential amino acids with structure.	02
1	15	In nucleic acid the sugar residue is in the furonose form. Justify.	02
1	16	What is DNA replication?	02
1	17	What are zwitter ions? Write the zwitter ionic form of glycine.	02
1	18	How does denaturation inproteins happens?	02
1	19	Name the hydrolysis products of nucleic acids.	02
1	20	Give an example for the action of CNBr on peptides.	02
2	21	Name any two saturated and unsaturated fatty acids. Mention their physical state.	02
2	22	What are reducing and non reducing sugars? Give examples.	02
2	23	What are reducing sugars? Give examples.	02
2	24	What are non reducing sugars? Give examples.	02
2	25	What are epimers? Give examples.	02

2	26	Define epimerization with example.	02
2	27	What is mutarotation?	02
2	28	Explain mutarotation?	02
2	29	Write the conformation structure of sucrose. Give its systematic name.	02
2	30	Mention the number of isomers possible for glucose?	02
2	31	Mention the number of isomers possible for	02
2	32	Mention the number of isomers possible for	02
2	33	Write the general formula of polysaccharide?	02
2	34	What are epimers? Mention the epimers of glucose?	02
2	35	Define sphingolipids?	02
2	36	What are Lecithins?	02
2	37	What are essential fatty acids? Give examples.	02
2	38	Mention the biological importance of oils and fats.	02
2	39	What are waxes?	02
2	40	What are trisaccharides? Give an example.	02
2	41	Name any two saturated and unsaturated fatty acids. Mention their physical state.	02
2	42	What are amylopectin?	02
2	43	What are homoglycans and heteroglycans?	02
2	44	What is aldotriose? Give example.	02
2	45	Define keto-sugars with two examples.	02
2	46	Draw the cyclic structure of α-D glucose and β-D glucose.	02
2	47	Draw the cyclic structure of α-D glucopyranoseand β-D glucofuranose.	02
2	48	Mention any two physiological importance of glycosides.	02
2	49	Define invert sugar.	02
2	50	What is cellobiose?	02
2	51	What are glycoprotiens? Give examples.	02
2	52	How many sterioisomers are possible for aldopentose?	02
2	53	How many sterioisomers are possible for ketohexose?	02
2	54	What are phospholipids? How are they classified?	02
2	55	Write the biological importance of lipids?	02
2	56	Write industrial application of cellulose.	02
2	57	Give the partial structure of starch.	02

2	58	Differentiate between the anomers and epimers.	02
2	59	Define Carbohydrates. Classify them with suitable examples.	02
2	60	Describe the structure and functions of Mucopolysaccharides.	02
2	61	Define polysaccharides. Classify them and write their importance.	02
2	62	Explain why sucrose is called as invert sugar	02
2	63	Define lipids and classify them with suitable examples.	02
2	64	Classify fatty acids in various ways with suitable examples.	02
2	65	How can we differentiate between a glycerophospholipid and a sphingophospholipid?	02
2	66	What are lipids? How are they classified?	02
2	67	What are prostaglandins? Give its one application.	02
1			
	68	Discuss the primary and secondary structures of proteins	05
1	69	Comment the structure and conformation of peptide bond	05
1	70	Explain the use of cyanogen bromide in the cleavage of polypeptides.	05
1	71	Write the steps involved in the synthesis of oxytocin.	05
1	72	Explain the use of Sanger's method in N-terminal determination of polypeptides.	05
1	73	Give steps involved in the formation of 3,5-phosphodiester bond by the DCC method.	05
1	74	Write a note on Crick-Watson model of DNA.	05
1	75	Illustrate the protocols employed for the protection of hydroxyl groups in the synthesis of polynucleotides.	05
1	76	Explain the role of nucleic acids in the biosynthesis of proteins.	05
1	77	Explain, in detail types of secondary structures of proteins.	05
1	78	Write a note on Denaturation of proteins.	05
1	79	Explain the secondary structure for the proteins.	05
1	80	How the amino acids are synthesized by strecker and Gabrilphthalimide method.	05
1	81	Discuss denaturation and renaturation of proteins.	05
1	82	Explain briefly on the protection of amino groups and carboxylic acid groups in peptide synthesis.	05
1	83	Describe the role of nucleic acids in the biosynthesis of proteins.	05
1	84	Describe the general methods available for determining the primary structure of proteins.	05
1	85	How is CNBr degradation useful in the structural elucidation of peptides?	05
1	86	Write a note on secondary structure of proteins.	05

1	87	Draw the structure of the bases present in DNA molecule and give a brief account of Crick-watson model of DNA.	05
1	88	Describe the solid phase synthesis of dipeptide.	05
1	89	How could thecarbobenzyloxy method be used to prepare a peptide.	05
1	90	Draw and describe Watson-Crick model of DNA.	05
1	91	How N-terminal residue is determined by different methods in peptides.	05
1	92	Write an account of solid phase synthesis of peptides.	05
1	93	Give a brief account of protection and deprotection of functional groups during peptide synthesis	05
1	94	Write a note on primary, secondary and tertiary structure of proteins.	05
1	95	Discuss the Merrifield-solid phase synthesis with suitable examples.	05
1	96	Discuss the biological significance of Vasopressin and oxytocin.	05
1	97	Discuss the determination of amino acids sequenced of proteins using Sanger's and Edman's method	05
1	98	Explain mechanistically the use of CNBr in fragmentation of proteins into polypeptides.	05
1	100	Write the steps involved in the synthesis of a dipeptide by Merrifield's solid phase method	05
1	101	Discuss the use of enzymes in the selective cleavage of polypeptides with examples.	05
2	102	Outline the synthesis of maltose.	05
2	103	Outline the steps involved in the Smith degradation of polysaccharides. Explain how it is useful in the structural elucidation of polysaccharides.	05
2	104	What are phospholipids? Give a general method for the synthesis of phospholipids.	05
2	105	How to obtain 2, 3, 4, 6-tetra-O-methyl-D-glucose from maltose?	05
2	106	Give the synthesis and industrial applications of sucrose.	05
2	107	Give the biological applications of lactose.	05
2	108	Describe the biosynthesis of carbohydrates?	05
2	109	Outline the synthesis of lipids.	05
2	110	What are prostaglandins? Write a note on classification and biological significance of prostaglandins.	05
2	111	Write the most stable conformer for, glucopyranose and glactopyranose.	05
2	112	Discuss the general methods employed for determining the structure of polysaccharides.	05
2	113	Using periodic acid, how do you establish the ring size for glucose?	05
2	114	Give the structure and explain the biological importance of agar-agar.	05
2	115	Structure and properties of starch.	05
2	116	Clinical importance of Inulin and Dextran.	05

2	117	Structure and functions of three biochemically important disaccharides.	05
2	118	Explain the structure of starch and glycogen.	05
2	119	Define phospholipids. Classify them with suitable examples and state their functions.	05
2	120	Classify fatty acids in various ways with suitable examples.	05
2	121	Define stereoisomer, optical isomer, epimer and enantiomer.	05
2	122	Explain enzymatic degradation of cellulose to glucose.	05
2	123	Differentiate between lipid, fat, oil and waxes.	05
2	124	Explain with examples the importance of glycolipids and phospholipids in the cells.	05
2	125	What is mutarotation? Give its mechanism.	05
2	126	Outline the synthesis of maltose.	05

Question Paper Pattern- Model Question Paper

QP CODE: 55206

St. Philomena's College (Autonomous) Mysore

III Semester M.Sc. Final Examination : December - 2019

Subject: ORGANIC CHEMISTRY

Title: Bio-Molecules (SC)

Time: 3 Hours **Max. Marks: 70**

PART – A

Answer ALL the questions: **10×2=20**

1. a. What are phospholipids? How they are classified?
   b. Write the biological importance of lipids.
   c. What are denaturation and renaturation of proteins? Explain with examples.
   d. What is DNA replication?
   e. In nucleic acid the sugar residue is in the furanose form – Justify.
   f. Write industrial applications of cellulose.
   g. Give the partial structure of starch.
   h. Differentiate between the anomers and epimers.
   i. Write the biological importance of vasopressin.
   j. What are essential amino acids? Write any two essential amino acids with structure.

PART – B

Answer FIVE of the following questions: **5×10=50**

2. a. Outline the synthesis of oxytocin. **05**
   b. Explain the DNA structure by Watson-Crick model. **05**
3. a. Discuss the enzymatic method of *N*-terminal residue determination. **03**
   b. Describe the biosynthesis of carbohydrates. **07**
4. a. Write the steps involved in the synthesis of tripeptides by Merrifield solid phase synthesis. **03**
   b. Write the secondary structure of proteins. **04**
   c. Write a note on recimization and use of HOBT in peptide synthesis. **03**

PTO

5.	a.	Outline the synthesis of Lipids.	05
	b.	What are prostaglandins? Write a note on classification and biological significance of prostaglandins.	05
6.	a.	Write a note on formation of Internuleotide bond using DCC.	06
	b.	Write the most stable conformer for: glucopyranose and glactopyranose.	04
7.	a.	Discuss the general methods employed for determining the structure of polysaccharides.	05
	b.	Write the synthesis of adenosine and uridine.	05
8.	a.	Using periodic acid, how you establish the ring size for glucose.	05
	b.	Give the structure and explain the biological importance of agar-agar.	05

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St. Philomena's College (Autonomous) Mysore
II Semester M.Sc. Makeup Examination September 2018
Subject: ORGANIC CHEMISTRY
Title: Bio-molecules (SC)

ime: 3 Hours Max Marks: 70

struction to the Candidates: ***Write equations and draw neat labeled diagrams wherever required.***

PART –A

Answer the following: 10×2=20

a. What are Zwitter ions? Write the Zwitter ionic form of glycine.

b. What are epimers? Give example.

c. Write the structure and mention the linkage in maltose.

d. How do you differentiate starch and glycogen.

e. Give any two methods for the protection of – OH group in sugar moiety.

f. What are aldose sugar? Give examples.

g. Give biological importance of PGE1.

h. What are lecithins? Give examples.

i. What are saturated and unsaturated oils?

j. How does denaturation in proteins happens?

PART –B

Answer any FIVE of the following: 5×10=50

2. a. Discuss the solid phase Merrifield synthesis with suitable example.

   b. Explain with suitable example the synthesis of nucleotide using DCC. 6+4

3. a. Discuss Watson-Crick model of DNA.

   b. Discuss biological significance of vasopressin and oxytocin. 5+5

PTO

4. a. Discuss the determination of amino acids sequenced of protein using Sanger's and Edman's reagent.

   b. Discuss the classification and constitution of prostaglandins.

5. a. State Hudson's rule and explain muta rotation.

   b. Explain the determination of ring size of monosaccharides.

6. a. Explain primary, secondary, tertiary and quaternary structure of protein.

   b. Discuss the biological importance of Insulin.

7. a. Write a note of confirmation and configuration of glucose.

   b. Discuss the chemical method of selective cleavage of polypeptide bonds.

8. a. Discuss the classification and purification of lipids.

   b. Discuss the biological importance of glycosides and amino sugars.

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